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Internet Protocol

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Chapter 1

Internet Protocol

1.1 Introduction

Network protocols are designed in layers or levels, where every level is responsible for a different matter of communications. A protocol stack is a combination of different protocols at various levels. TCP/IP is considered a 4-level (or 4-layer) system as shown in Figure 1.1.

TCP and IP are two protocols that give name to an architecture network. However, a TCP/IP network also needs other protocols to have all the features. Figure 1.2 shows the protocols that make up the TCP/IP family.

The target of IP protocol is to turn physically different networks (such as Ethernet, Token Ring, X.25, Frame Relay, ATM ...) into an apparently homogeneous network, which is known as internet working. This way, IP hides underlying physical networks creating over them a single "virtual" network or a network of networks, wherein:

- There is an scheme of identification (or addressing) of every system, and it is uniform and universal. This addressing scheme has to be independent of *hardware*. These properties are achieved by assigning to each node a unique number of 32 bits (in IPv4, Internet Protocol Version 4) called IP address.
- Data travels through the virtual IP network grouped into packets called datagrams. The manipulation of these datagrams by the network is a standard method called "datagram routing" which is independent of the particular network where these datagrams are.
- The delivery of datagrams from one extreme to another over the IP network is:
  - Unreliable (there is no error control or flow control).
  - It is not connection oriented
  - The network does its best to deliver packets, however delivery is not assured (best effort).

Therefore datagrams may be lost, disrupt or even double, and IP will not consider these situations (they will be controlled by protocols above). That is, the IP protocol assumes no high things from the lower layers. Only that datagrams "probably" will be transported to the host destination.

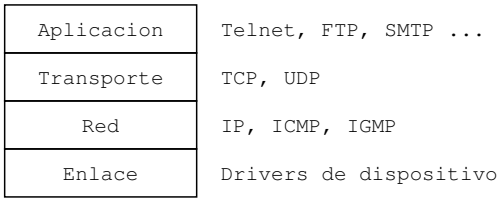


Figure 1.1: The 4 layers of TCP/IP

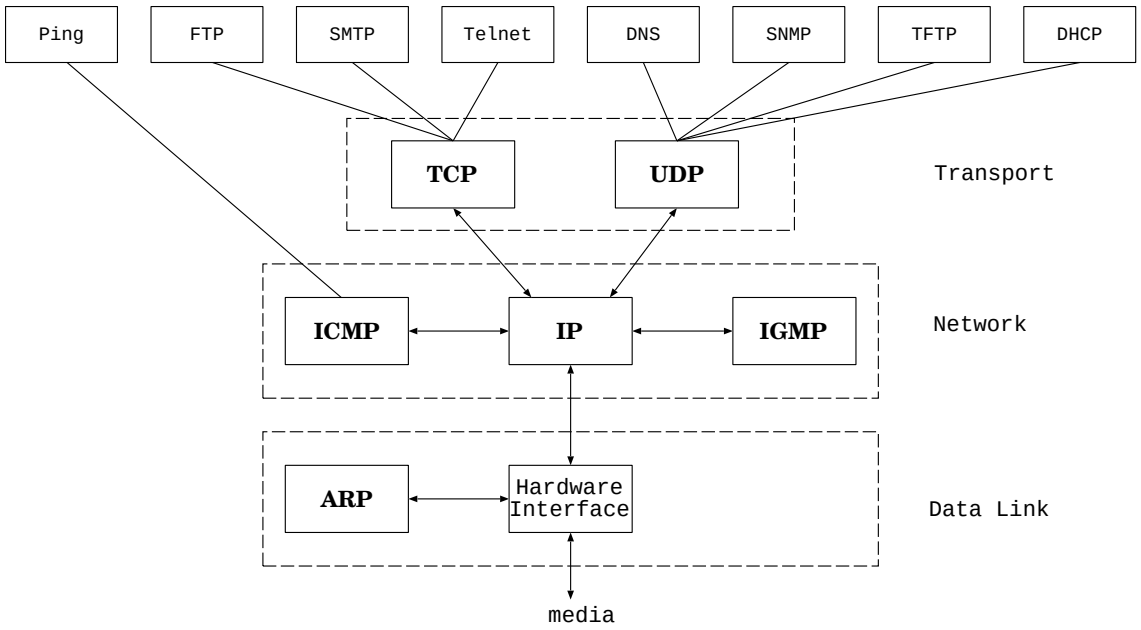


Figure 1.2: Protocols of the TCP/IP family

1.2 Internet addresses

Internet addresses can be symbolic or numeric:

- Symbolic addresses are those that humans usually use because they are easier to remember than numeric. These addresses are character strings separated by dots. For example: **www.upc.es**.
- The numeric addresses are commonly used by the software IP. The numeric form is a sequence of 32 bits. To be more intelligible, it is usually expressed as decimal numbers separated by dots (dotted-decimal), where each decimal number represents 8 bits (range 0-255). For example, the binary address **10010011. 01010011. 00010100. 00000010** can be expressed in decimal notation as **147.83.20.2** (which in turn corresponds with the symbolic address **www.upc.es**).

Mapping functions between the symbolic addresses and numeric addresses are made by the DNS (Domain Name System), as seen in previous practices.

An address does not identify a device on the Internet, but a certain network interface of the device. In other words, the IP address identifies a particular machine in a particular physical network. When the machine is connected to more than one network, the machine is called "*multi-homed*" and it has an address for each network interface.

Indeed, in order to correctly interpret a particular IP address, a second sequence of 32 bits called “mask” (netmask) is required. The mask is used to divide the address in two parts:

- The network part, formally called “network number” (RFC 1166), although sometimes we use the terms network address and netID. Network numbers are centrally managed by the INTERNIC (INTERnet Network Information Center) and they are unique throughout the whole Internet. When someone asks an IP addres to the INTERNIC, no address is assigned to each individual machine, but there is a network number given, allowing to assign, as needed, all the IP addresses which are valid within this range. The host part, formally called “host number”, although occasionally the terms host address and hostID are used.

In order to divide an IP address in network number and host number the mask is used. So, those bits of the address where the mask has a "1" are part of the network number, while the positions where the mask has “ 0” are part of

host number. For example in the next address 10010011. 01010011. 00010100. 00000010 with a mask 11111111. 11111111. 00000000. 00000000 we have a network number 10010011. 01010011. and a host number 00010100. 00000010.

By convention the network number is in the most significant bits (the bits on the left of the address) while the host number is usually in the least significant bits (the bits on the right of the address). Additionally the network and host numbers usually consist of consecutive bits of the IP address (it is very uncommon to see masks with non-consecutive zeros and ones).

The network mask is also commonly represented in decimal format so that for the previous example we have the address 147.83.20.2 and the mask 255.255.0.0. Another way to represent an IP address is setting the address in decimal format and the mask as a decimal number that indicates how many consecutive ones form the number network. So for our example, the address 147.83.20.2 with mask 255.255.0.0 can be written as 147.83.20.2/16.

The size of the part devoted to host depends on the size of the network. To meet multiple needs some different network classes have been defined, depending on the place where the IP direction is divided. This way, we will have the following classes:

Class A

Class A covers networks from 1.0.0.0 through 127.0.0.0. The network identifier is contained in the first octet, leaving for the host the remaining 24 bits, allowing approximately 1.6 million of machines per network. Therefore, the resulting network mask is 255.0.0.0 (11111111. 00000000. 00000000. 00000000 in binary format). In this class, the range 10.0.0.0 to 10.255.255.255 is a reserved range for private use. A private host address (or private e-host) would be 10.0.0.1 / 8.

Class B

Class B contains networks from 128.0.0.0 to 191.255.0.0, where the netid is in the first two octets. This class allows 16320 networks with 65024 places each network. The mask of this class is 255.255.0.0 (11111111. 11111111. 00000000. 00000000 in binary). In this class the range 172.16.0.0 to 172.31.0.0 is reserved for private use. A private e-host (or privatehost address) can be 172.31.0.1/16.

Class C

Class C networks range from 192.0.0.0 to 223.255.255.0, where the netid is in the first three octets. This class allows nearly 2 million networks with 254 locations each network. The mask of this class is 255.255.255.0 (11111111. 11111111. 11111111. 00000000 in binary). In this class the range 192.168.0.0 to 192.168.255.0 is reserved for private use. One possible private host address can be 192.168.128.1/24.

Class D

Addresses in class D are addresses within the range 224.0.0.0 to 239.255.255.255 and they are called multicast addresses.

Class E

Addresses in class E are addresses within the range 240.0.0.0 to 247.255.255.255. They are either experimental classes or they are reserved to future applications.

Special addresses

Note that in the part reserved for host, the values 0 and 255 are reserved because they have a special meaning:

- An IP address with only zeros in the host part refers to the network address itself.
- An address with only ones in the host part is called broadcast address as it refers to all the hosts on the specific network. So 147.83.20.255 is not a valid hostid, but it refers to all the hostid on the 147.83.20.0 network.

Network addresses 0.0.0.0 and 127.0.0.0 are reserved. The first one is called the default route (where IP routes datagrams), and the second one is the loopback address.

Network 127.0.0.0 is reserved for local traffic IP of the machine. Typically the address 127.0.0.1 is assigned to an interface of the machine (loopback interface) which acts as a closed circuit. Any IP packet sent to this interface will be returned as if it had returned from a network. This allows, for example, to install network software or to test ethernet cards, even in the absence of a real network.

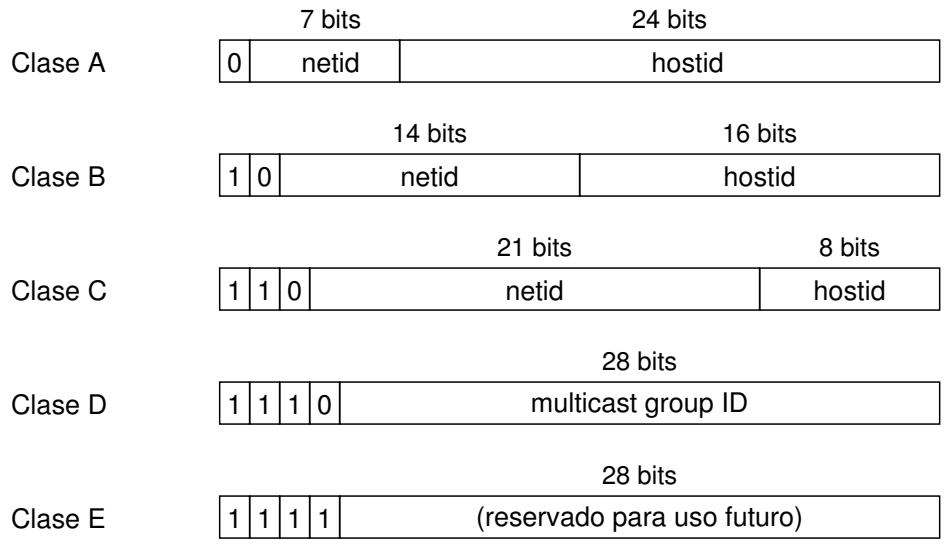


Figure 1.3: IP address classes

1.3 IP datagram

The IP datagram is the unit of transfer in IP networks. It is, basically, an IP header and a data field intended to upper protocols. The IP datagram is encapsulated in the link-level frame, which usually has a maximum length (MTU, Maximum Transfer Unit), depending on the network hardware used. For Ethernet, MTU is typically 1500 bytes. Rather than limiting the datagram to a maximum size, IP protocol can treat the fragmentation and the reassembly of the datagrams. In particular, IP does not impose a maximum size, but it states that all the networks should be able to handle at least 576 bytes. Every datagram fragment has a header, basically copied from the original datagram, and from the corresponding data. The fragments are treated as usual datagrams while they are being transported to their destination. Note, however, that if one of the fragments is lost, the entire datagram is considered lost, and the remaining fragments will be also considered as lost.

1.3.1 IP datagram format

The IP datagram header consists of the fields displayed in Figure 1.4.

Wherein:

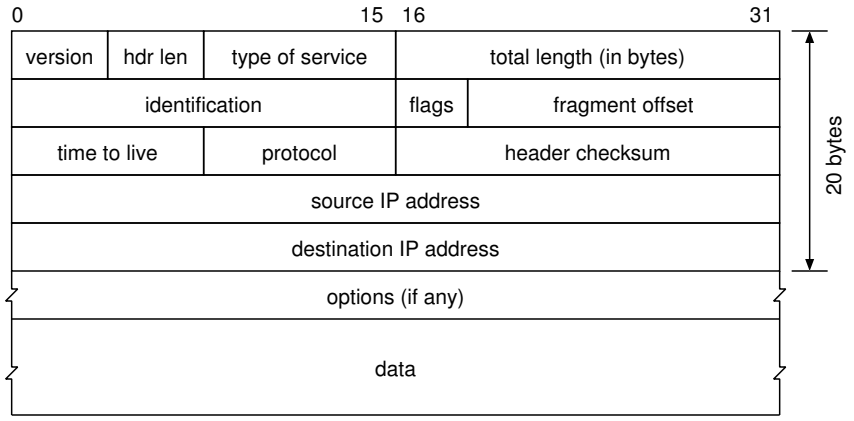


Figure 1.4: IP datagram format

Version is the IP protocol version. The current one is version number 4. The number 5 is experimental and 6 is IPng.

Hdr Len is the IP header length counted in 32-bit amounts. It does not include the data field.

Type Of Service is the type of service. It is a quality indication of the requested service for this IP datagram. A detailed description of this field can be found in RFC 1349.

Total Length is the total length of the datagram, header and data, specified in bytes.

Identification is a unique number assigned by the issuer to help reassembling a fragmented datagram. Fragments of a datagram have the same identification number.

Flags are flags to fragmentation control.

Wherein:

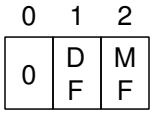


Figure 1.5: Detail of the Flags field

Bit 0 is reserved and must be 0.
Bit **DF** means **D**o not **F**ragment. Fragmentation is allowed with 0 but it is not with 1.
Bit **MF** means that there are **M**ore **F**ragments. 0 means that it is the last fragment of the datagram and 1 means that there are more fragments.

Fragment Offset (FO) is used in fragmented datagrams in order to help the reassembly of the entire datagram. The value is the number of 64 bits amounts (the header bytes are not included) contained in previous fragments. In the first (or only) fragment, FO value is always zero.

Time To Live specifies the time (in seconds) that this datagram is allowed to travel. Every "router" this datagram passes by must substract from this field the time that this "router" takes to process the datagram. Actually, a "router" can process a datagram in less than 1 second; hence the "router" will substract one from this field and so, the TTL becomes an account of hops rather than in a time metric. When the value reaches zero, it is assumed that this datagram has been traveling in a loop and is discarded. The initial value should be fixed by the higher-level protocol that creates the datagram.

Protocol shows the official number of the higher-level protocol to which IP should deliver the data of the datagram. Some significant values are shown in Table 1.1.

Protocol	Number
Reserved	0
ICMP	1
IGMP	2
IP encapsulated	4
TCP	6
UDP	17
OSPF	89

Table 1.1: Some Protocol Numbers

Header Checksum is calculated as the complement to one of the sum of the complements to one of all the words of 16 bits of the header. If the checksum header does not match the contents, the datagram is discarded, since at least one bit of the header is corrupted, and the datagram might have reached a wrong destination.

Source IP Address is the IP address of 32 bits of the sender host.

Destination IP Address is the IP address of 32 bits of the receiver host.

Options is a variable length field. The options are, in principle, included for testing or debugging networks, therefore it is not required that every IP implementation is capable of generating options in the datagrams that the same IP implementation creates, but it is expected to be able to process datagrams containing options.
The field **Options** has a variable length depending on the selected option. Some options have a length of a single byte and others have variable lengths. The first byte of any option is called type byte and its structure is shown in Figure 1.6.

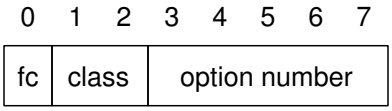


Figure 1.6: Type byte detail

Wherein:

FC is the **F**lag **C**opy and it indicates whether the field has options to copy (1) or not (0) when the datagram is fragmented.

class is a 2 bits unsigned integer, wherein:

Option class	meaning
0	Network control or datagram
1	Reserved for future use
2	Debugging and measurement
3	Reserved for future use

Table 1.2: Option classes

number is a 5 bits unsigned integer which indicates the choice number within each class.

1.3.2 The IP datagrams fragmentation

When an IP datagram travels from one host to another can cross different physical networks. The physical networks impose a maximum size frame, called MTU (Maximum Transmission Unit), which limits the length of a datagram.

Therefore, there is a mechanism for fragmenting large IP datagrams into smaller ones, and then reassembling them in the destination host destination. IP requires that each link has an MTU of at least 68 bytes, so that if any network provides a lower value, the fragmentation and reassembly will be implemented in the network interface layer transparently to IP. Note that 68 is the sum of the largest header IP, 60 bytes, and the minimum size of the data in a fragment (8 bytes).

IP implementations are not required to handle unfragmented datagrams over 576 bytes, but most of the IP implementations can handle larger values. Typically above 8192 bytes or even more, and rarely less than 1500.

An unfragmented datagram has only zeros in its fragmentation information. That is, the flags **FC** and **FO** are 0.

When the fragmentation has to be performed, the following steps are executed:

- It is checked the **DF** flag bit to see if fragmentation is allowed. If it is set to one, the datagram is discarded and an error is returned to the sender using ICMP[1](Internet Control Message Protocol).
- Based on the MTU value, the data field is divided into parts where each part, except the last one, must be multiple of 8 bytes.
- Next, these portions of data are placed in IP datagrams.
- Headers are copied from the original header, with some modifications:
  - The **MF** flag bit is set to one in all fragments except the last one.
  - The **FO** field is set to the value of the location of the corresponding data portion.
  - If options were included in the original datagram, the **FC** bit of the type byte determines whether or not the data will be copied in all fragments or only in the first one[2][3].
  - **Hdr Len** field (header length) is initialized.
  - **Total Length** field (total length) is initialized.
  - **Header Checksum** of the header is recalculated.

Each one of these datagrams is sent as an usual IP datagram. IP handles each fragment independently, ie, fragments can pass through various routes to their destination, and they may be subject to further fragmentations if they pass networks with lower MTUs.

In the destination host, the data must be reassembled. For this purpose the following steps are followed:

- To reassemble the fragments, the receiver allocates a storage buffer as the first fragment has arrived.
- once the first fragment has arrived, a temporary counter starts[4].
- Each fragment is identified by the **Identification** field which is a number unique within the limits imposed by the use of a 16 bit number. Given that the fragmentation does not change this field, as fragments arrive at the destination they can be identified with this identifier and the IP addresses of the source and destination of the datagram. Besides, the **Protocol** is also checked.
- The arriving fragments are copied to the buffer at the location indicated by the **FO** field.

[1] See the section 1.7, dedicated to this protocol.

[2] Por ejemplo, las opciones de encaminamiento de la fuente se tendrán que copiar en todos los fragmentos y por tanto tendrán a uno este bit.

[3] For example, the routing options of the source shall be copied into all fragments and thus they will be set to one in this bit.

[4] In fact, the process of reassembly is not a task of the IP layer. This process depends on the upper layer (transport layer). So, the transport layer is the layer in charge of implementing this counter and realizing the loss of fragments. This layer could be TCP to transport in a connection-oriented network or UDP in connectionless-oriented networks. The value of this counter is often the **TTL** field, although some implementations allow to configure the counter.

- When all fragments have arrived, the original datagram is restored and it continues its process.
- If the temporary timer expires and not all the fragments have been received, the datagram in the reassembly phase is discarded.

For a full understanding of the options listed below, a prior reading of paragraph 1.6 is recommended.

Record Route. RR

This option provides a way to store the IP addresses of the routers through which the datagram is routed. The source host is responsible for setting the option, saving space enough to store the IP addresses of the routers through which the datagram passe by. If the space to storage the route is filled before the datagram reaches its destination, the datagram is broadcast, but the recording of the route will be stopped.

Figure 1.7 shows the format of this option:

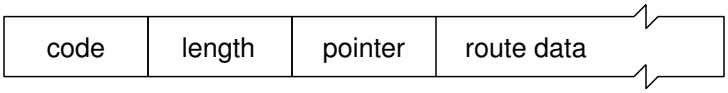


Figure 1.7: Format of the RR option

Where:

Code contains the class and the number of RR option (**Code** has a decimal value 7 for this option).

Lenght indicates the total length in bytes of the option.

Pointer is used to point to the next available slot to store an IP address. This is a field in bytes, so each router enters its IP address into the appropriate slot and increases **Pointer** at 4. If at any router happens that **Pointer> Lenght,** then the recording of the route will be stopped from that router on.

IP Address *n* indicates the address registered by the router *n.*

Source Route or SR)

This option gives to the transmitter a thinner way to control the routing process which is usually based on a single destination address. With this option, the issuer may provide information about the route to be followed by datagram[5].

IP provides to different ways to Source Routing:

- Strict Source Routing or SSR.
  The source provides the "strict" path by which datagrams should be directed to the destination. When we say "strict" we mean that the path between two successive addresses from the list should consist of a single physical network. If a router can not route a datagram to the following address from the list, an error happens and an ICMP message is generated indicating unreachable destination (see section **??**).
- Loose Source Routing or LSR.
  In this case the list specified by the issuer is not strict, that is, it is allowed to make multiple physical networks hops between successive divisions from the list.

The format is equivalent to the RR option (Figure **??**) but the performance is a bit different:

- For SSR and LSR the **Code** field are worth 137 and 131 respectively.
- The source host puts in the IP Destination field the address of the first router of the path.

[5]This option can route datagrams through a particular physical network and thus, status can be checked, maintenance tasks can be done and so on.

- The *n* **IP Address** fields store the IP addresses of the routers of the path.
- When a datagram reaches its next destination and the source route is not empty (**pointer < lenght**) the receiver:
  - Takes the address in the **IP Address** field given by **pointer** and places it as the new destination address of the datagram.
  - Replaces the address in the **IP Address** field indicated by **pointer** for its local IP address in the network that will be used to send the datagram.
  - Increases **pointer** in 4 (ie 32 bits).
  - Transmits the datagram to the new IP address destination.
  - This procedure ensures that the return route is recorded in reverse order so that the receiver may use data from the option to build the way back.

Internet Timestamp or IT ()

The timestamp is an option to force some (or all) of the routers on the path to the destination to put a time stamp timestamp in the data of the option. The timestamps are measured in seconds and they can be used for debugging. They can not be used to measure the network performance because of two reasons:

1. They are not precise enough, because most of the datagrams are sent in less than one second.
2. They are not precise enough because the routers usually do not have their clocks synchronized.

1.4 Address resolution (ARP)

In a single physical network, the individual hosts are known in the network through their physical addresses or addresses of level 2. The high level protocols route the datagrams to the destination by global addresses or addresses of level 3 (in the case of IP network addresses it is an IP address) of the virtual network. The drivers of the network device of the physical network (eg the drivers of the Ethernet cards for PCs in the lab) do not understand the addresses of level-3.

Is therefore necessary to perform an address translation from level 3 to addresses of level 2. In the case of IP networks over Ethernet the translation is performed from IP address to Ethernet address (which is a sequence of 48 bits). This translation is done in the host using a table usually called "ARP cache"[6]. When the address is not in the ARP cache the host uses the ARP protocol (Address Resolution Protocol) to find it out.

ARP is a request/response protocol. The request is sent in a shape of a level 2 broadcast network. If one of the machines of the network recognizes its own IP address in the request, it returns an ARP response to the host who requested it. The response will contain the physical address corresponding to the requested IP address. This address will be stored in the ARP cache of the applicant host. All subsequent sent datagrams to this IP address may be associated to the corresponding physical address, which will be used by the driverof the network device to send the datagram.Figure 1.8 shows the ARP frame format.

If an application wants to send data to a particular IP address destination, the IP routing software first determines the next hop IP address (which can be its own destination host or a router) and secondly the network device that data should be sent. Next the ARP module is consulted to find the physical address.

The ARP module tries to find the address in its cache. If found, the ARP module returns the corresponding physical address. If not, the ARP module discards the packet (it is assumed that being a high level protocol it re-submit it) and generates a network broadcast message in order to request an ARP for this address.

When a host receives an ARP packet (either a broadcast or an answer point to point), the receiving device passes the packet to the ARP module, which will add this packet to the ARP cache. The next time a higher level protocol wants to send a packet to that host, the ARP module will find the hardware address to send the package to.

[6]To query and manipulate the ARP cache of a Linux, the `arp` command is used. For more information on this command see the arp man page.

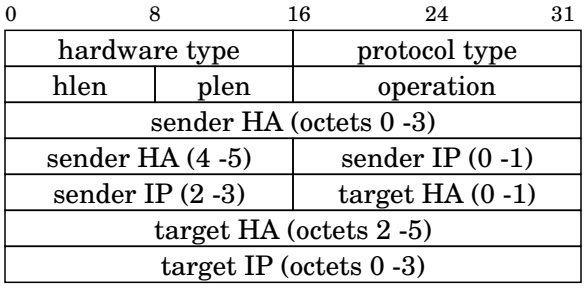


Figure 1.8: ARP frame format

1.5 Subnets

Obviously, a class A address is only assigned to networks with a large number of hosts, and that addresses in class C are suitable for networks with few hosts. However, this means that medium-sized networks (those with more than 254 hosts or where it is expected that it will have more than 254 hosts in the future) must use Class B addresses. The number of networks of small and medium size has been growing very quickly in recent years and it was feared that, if it had allowed this growth was maintained, all Class B addresses would have been used by the mid-the 90's. This is what is known as "the problem of exhaustion IP addresses".

As discussed in section 1.2 each part of the IP address (network number and number of host) is the responsibility of different organizations: the network number is assigned by the ITERNIC, while the allocation of host numbers is is the responsibility of the authority that controls the network. As outlined below the host number can be further divided: this division is also controlled by the authority which owns the network, not INTERNIC.

Due to the explosive growth of Internet,the use of IP addresses assigned by standard classes became too rigid to allow easily change the configuration of local networks. These changes might be needed when:

- A new physical network is installed.
- The growing of the number of hosts requires splitting the local network into two or more networks.

To avoid having to request additional IP addresses in these cases, the concept of subnet was introduced. Ins subnets the host number of the IP address is subdivided again into a number and host network. This second network is called subnet. The main network now consists of a set of subnets and the IP address is interpreted as

<network number, subnet number, host number>

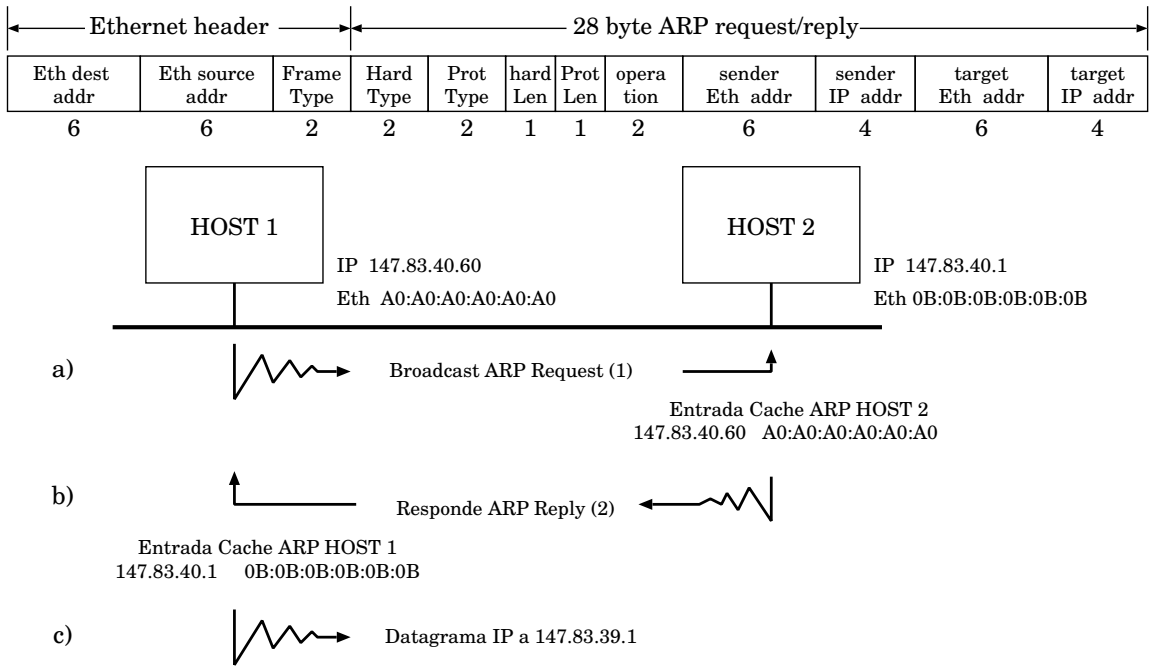
The combination of the subnet and host number is often called "local address" or "local part". Subnetting is implemented in a way that is transparent to remote networks. A host within a subnetted network is aware of their existence, but a host in a different network is not, and still considers to the local IP address as a number of host.

The division of the local IP address into subnet and host numbers is a free choice of the local administrator. Any series of bits of the local part can be taken to the subnet required. For example, the subnetting is widely used by ISPs (Internet Service Provider) that are responsible for providing service to Internet access, they tend to be assigned a public addresses Internet range, which are divided into subnets and then they are assigned to their customers.

The division is performed using a netmask other than those that divide the IP addresses in standard classes. The operation of these masks is equivalent to that described in section 1.2 to the standard classes, ie, the bits to zero in this mask indicate positions of bits for the host number, and those set to one indicate positions of bits for the subnet number. The positions of the mask belonging to the network number are set to one but not used.

The special treatment of "all bits to zero" and " all bits to one" is applied to each of the three parts of subnetted IP address just as an IP address without subnets. Note that when subnetting, the number of machines on which it is broadcast is reduced, optimizing the useful traffic within the network.

As an example suppose we have the private class C network 192.168.1.0/24. In this case we have a total of 254 (ie $2^8 - 2$, having 8 bits of hostid) possible addresses of host, from 192.168.1.1/24 to 192.168.1.254/24. Note that



(1) ARP Request

FF:FF:FF:FF:FF:FF-A0:A0:A0:A0:A0:A0-0806-0001-0800-06-04-01-A0:A0:A0:A0:A0:A0-147.83.40.60-00:00:00:00:00:00-147.83.40.1

(2) ARP Reply

A0:A0:A0:A0:A0:A0-0B:0B:0B:0B:0B:0B-0806-0001-0800-06-04-02-0B:0B:0B:0B:0B:0B-147.83.40.1-A0:A0:A0:A0:A0:A0-147.83.40.60

Figure 1.9: Ejemplo de ARP

Figure 1.10: ARP example

addresses 192.168.1.0 and 192.168.1.255 are reserved. The first one to identify the network and the second one as the broadcast address. Suppose as well we have divided this address into two subnets of the same dimension, since we have two offices in different populations and suppose we will assemble two LANs connected by WAN (Wide Area Network transport network rented to any operator). Thus, we have that the hostid part has to give ground for subnet part. Since we need to do two subnets, a bit is enough to identify the subnet on which we are, staying two subnets: 192.168.1.0/25 and 192.168.1.128/25 with 126 possible host addresses in each subnet (ie $2^7 - 2$, with 7 for hostid. We summarize the network, mask and hosts addresses involved in the subnetting of this example.

Original basic networking: 11000000. 10101000. 00000001. 00000000 = 192.168.1.0/24
Mask: 11111111. 11111111. 11111111. 00000000 = 255.255.255.0 = /24

There are 254 possible addresses to assign to the network:
host # 0 11000000. 10101000. 00000001. 00000001 = 192.168.1.1/24 (*netmask* 255.255.255.0)
host # 1 11000000. 10101000. 00000001. 00000010 = 192.168.1.2/24 (*netmask* 255.255.255.0)
host # 253 11000000. 10101000. 00000001. 11111110 = 192.168.1.254/24 (*netmask* 255.255.255.0)

Doing subnetting, this two subnets will stay:
Subnet # 0 11000000. 10101000. 00000001. 00000000 = 192.168.1.0/25 (*netmask* 255.255.255.128)
Subnet # 1 11000000. 10101000. 00000001. 10000000 = 192.168.1.128/25 (*netmask* 255.255.255.128)

With the following host addresses available in each office:

Subnet # 0 for an office:
host # 0 11000000. 10101000. 00000001. 00000001 = 192.168.1.1/25 (*netmask* 255.255.255.128)
host # 1 11000000. 10101000. 00000001. 00000010 = 192.168.1.2/25 (*netmask* 255.255.255.128)
host # 125 11000000. 10101000. 00000001. 01111110 = 192.168.1.126 / 25 (*netmask* 255.255.255.128)
Subnet # 1 for the other office:
host # 0 11000000. 10101000. 00000001. 10000001 = 192.168.1.129/25 (*netmask* 255.255.255.128)
host # 1 11000000. 10101000. 00000001. 10000010 = 192.168.1.130/25 (*netmask* 255.255.255.128)
host # 125 11000000. 10101000. 00000001. 11111110 = 192.168.1.254/25 (*netmask* 255.255.255.128)

1.6 IP Routing

1.6.1 Introduction

The most important function of the IP layer is the datagram routing end to end through the virtual network. Therefore, this feature provides the mechanisms needed to interconnect different physical networks. The formation of the virtual network that connects multiple networks is achieved by means of some special hosts called "routers".

It is important to distinguish between a hub, a bridge, a router and a gateway.

The hub interconnects LAN segments at the network interface level and it sends frames between them. The hub serves as an extension of the physical cable that connects the machines on the LAN and its only function is to spread the signal coming from a certain port (input) to other ports. Hubs can be passive (if they do not amplify the signals received by its ports) or active (if they do amplifiy).

The switch is a device similar to the hub but now the switch performs switching between different ports, i.e. it switches packets observing their physical origin/destination addresses.

The Bridge interconnects LAN segments at the network interface level and sends frames between them. A bridge performs the MAC (Medium Access Control) relay function and it is independent of any higher layer (including LLC). A bridge provides, if needed, conversion of MAC level protocols. It is transparent to IP. That is, when a host sends a datagram to another host in a network which is connected via a bridge, it sends the datagram to the host and the datagram crosses the bridge without the sender knowing. The bridge is capable of learning hardware addresses of the machines it has in each port and it is able to isolate traffic and collisions of each LAN segment.

The router interconnects different physical networks at the network layer and it routes packets between them. The router must understand the addressing structure associated with the protocols it supports (IP in our case) and must choose the best transmission paths and optimal sizes for datagram, performing fragmentation if appropiate.

The gateway interconnects networks at higher levels than bridges and routers. A gateway usually holds address mapping from one network to another, and the conversion of data between different environments to achieve connectivity between the ends of the communication. Gateways typically provide two networks connectivity for a subset of application protocols supported on each of these two networks. A gateway is opaque to IP. That is, a host can not send an IP datagram through a gateway: it can only send it to the gateway. The gateway is responsible for transmission to the other network with the information from the higher level protocols that go into it.

Closely linked to the concept of gateway it is the **firewall** concept, which is used to restrict the flow of datagrams between networks for security reasons.

1.6.2 Direct and indirect delivery

The IP routing can be divided into two parts: direct delivery and indirect delivery:

- The direct delivery is the transmission of a datagram from the origin host origin to the destination host through a single physical network, in other words, two hosts can only communicate through direct delivery if both are directly connected to the same physical network (for example, a single Ethernet network).
  Basically in the direct delivery sender encapsulates the datagram in a physical frame, transforms the destination IP address into a physical address and sends the resulting frame to the destination via the corresponding hardware device driver.

- The indirect delivery is required when the destination host is not directly connected to the origin network, which means that the datagram must traverse multiple physical networks, and this requires cross routers.
  The indirect delivery is more complex given that the origin host has to identify the router that should receive the datagram. The first router must identify the next router that will receive the datagram. It is also called "to identify the next hop".
  The communication between two consecutive routers on the path is always done by direct delivery, ie a given router of the path and the nexthop router must be connected to the same physical network, but can not be notified. In turn, the last router of the path followed by the datagram must be connected to the same physical network as the destination host.

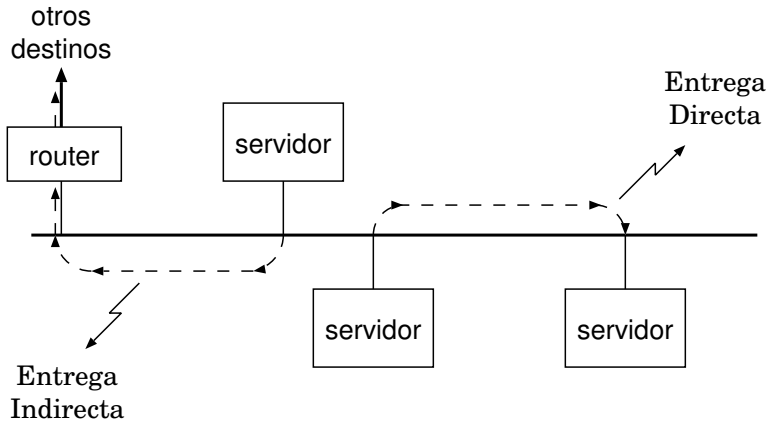


Figure 1.11: Direct and indirect delivery

The origin host finds out whether to perform directly or not, ie if destination host is directly connected or not to its physical network using the network prefix. The origin host extracts the network number of the destination IP address and compares it to the network number of its own IP address. If both match means that direct delivery can be used, if not, the origin host will perform indirect delivery

1.6.3 The routing algorithm

The method used by a router or a host to find the next machine to receive a given datagram is generically known as "routing algorithm".

The vast majority of routing algorithms use "routing tables". In the routing tables of every host or router the information about possible destinations is stored as well as how to achieve them. The information contained in routing tables should be minimal, since if each routing table had information about each possible destination address, it would be impossible to keep tables updated. In addition, the machines would not have enough space or processing power to handle them.

The goal is to minimize the information that the tables must keep by applying a scheme of global information hiding, keeping only the minimum necessary local information. Fortunately, the IP addressing scheme allows to do this easily: as mentioned in Section 1.2, the IP address is divided into network number and host number. Through this scheme it is possible to store network numbers in routing tables instead of full IP addresses. This way, some details

are hidden: which hosts and which way are connected to the different networks and its also the size of the routing tables is minimized.

The contents of the routing tables are often <N,R> pairs. Where N is a network number and R is the IP address of the router in the next hop to achieve such a network (hence the router must be connected to the same physical network).

To further simplify routing tables, shows the concept of "default route" appears. The default route contains the address of the next-hop router to which datagrams should be sent (also called default router) if after touring the routing table no specific route was found for the network number to which the datagram is addressed.

Although the desirability of routing based on destination network number has been already mentioned, the routing table allows to specify a special path for a particular host.

This way the basic algorithm routing of an IP datagram is as follows:

1. Extract IP destination addres D.
2. Compute the network prefix N with local mask.
3. If N corresponds to a physical network to which we are connected, a direct delivery is made (performing ARP).
4. If a direct delivery is not possible, it checks for a specific route for D and if so it sends the datagram to the next hop router specified in the table.
5. If there is no specific route, it checks for a route for the network N and if so it sends the datagram to the next hop router specified in the table.
6. If there is no route for N, the datagram is sent to the default router specified in the table.
7. If there is no default route and the IP routing software has reached this point, an error occurs (which can be reported by ICMP).

As important conclusion of the IP routing algorithm can be noted that:

1. In most implementations, the traffic targeted to a given network from an origin host will follow the same path even though there are various possibilities.
2. Only the last router of the path can determine whether the destination host is available (these situations are reported by ICMP). In addition we also need reports from the intermediate routers if any problem occurs.
3. Datagrams traveling from A to B may follow different routes to travel to datagrams travelling from B to A.

Figure 1.12 shows 4 networks interconnected via two routers and the routing table of the router R_1.

It is important to understand that except for the TTL field decrease, the routing software does not change the routing header from original datagram. In particular, the source and destination IP addresses remain unchanged during the entire route[7].

With respect to incoming datagrams:

- When a datagram arrives at a host the network devide driver delivers it to the IP software for processing. The IP software determines if the datagram is for the host itself, in which case the datagram goes to the high level protocol appropiate. The datagram is discarded if it is not for the host itself.
- In the case of routers, they must route the datagram if it is not directed to them.

Decide whether or not a machine is the recipient of a datagram is not a trivial task as it may seem at first glance. First there may be many network interfaces each with its corresponding IP address, the corresponding subnet ID (if the network is divided) must be checked and also it must recognize the broadcast messages and those on the multicast.

[7]Except in SR option discussed in section 1.3.2

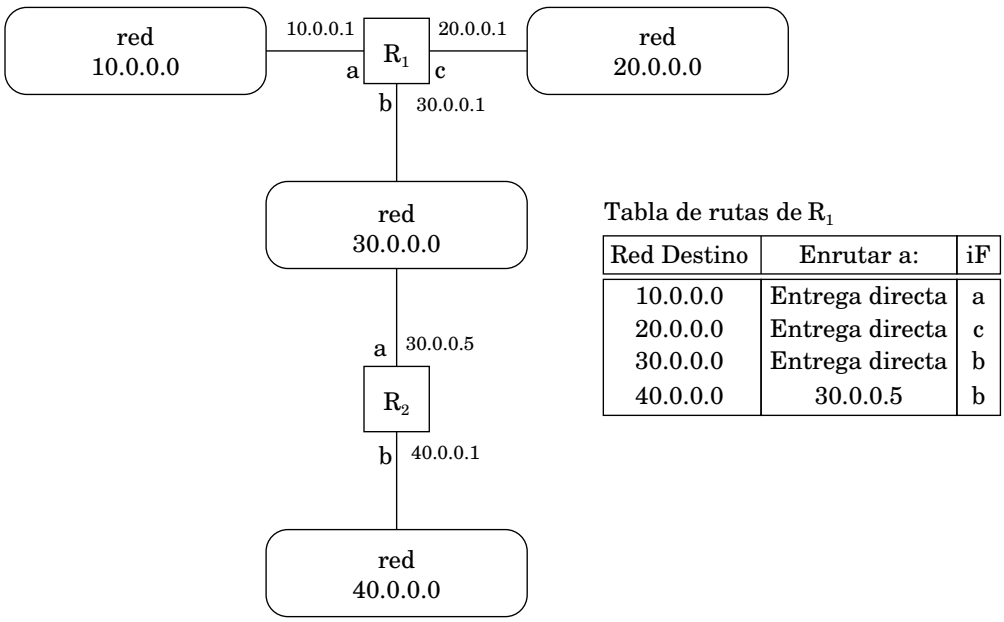


Tabla de rutas de R_1

Red Destino	Enrutar a:	iF
10.0.0.0	Entrega directa	a
20.0.0.0	Entrega directa	c
30.0.0.0	Entrega directa	b
40.0.0.0	30.0.0.5	b

Figure 1.12: Routing example

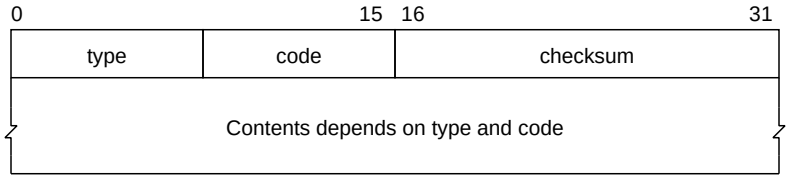


Figure 1.13: ICMP message

1.7 ICMP- Internet Control Message Protocol

ICMP (Internet Control Message Protocol) is a protocol used to signal errors or other conditions that may occur during a communication. These messages can be generated by any network device of level 3 that needs to report an error condition to a sending device whose transmission has caused the error. This condition implies that ICMP messages must be encapsulated in IP datagrams so that they can be routed from anywhere on the network to the receiver of the ICMP message. However, as shown in Figure 1.2, ICMP does not use any transport protocol as the ICMP message is not addressed to the communication application that generated the error, but the TCP/IP protocol stack will handle the ICMP message and will act accordingly (for instance, indicating the error condition to the application). An ICMP message has the structure shown in Figure 1.13.

Both the type and code fields determine the different ICMP messages and the format of the message content. Table 1.3 shows different types of ICMP messages classified by the type and code fields.

It is important to note that many of the ICMP messages are obsolete, or some of them are implemented in hosts and others in routers, or some of them are specifically disabled because they have proven to be security weaknesses .

Table 1.3 shows different types of ICMP messages. The last two columns in the table indicate that there are two types of ICMP messages: requests (query) and errors (query) ICMP error messages are messages that are sent to the device which packet generated the error. In this case, the ICMP error messages include the IP header of the datagram that generated the error (20 bytes) and the first 8 bytes of the datagram (including the start of the TCP header or UDP). Thus, the device that generated the error knows which application is responsible for the error generated.

ICMP query messages are used to request specific information (depending on the ICMP type and code) of a network device (hosts, routers, . . .). The query message corresponds to those that are labeled with request at the table,

type	*code*	*Description*	*Query*	*Error*
0	0	echo reply	•	
3		destination unreachable		
	0	network unreachable		•
	1	host unreachable		•
	2	protocol unreachable		•
	3	port unreachable		•
	4	fragmentation neepepitoded but fg not set		•
	5	source route failed		•
	6	destination network unknown		•
	7	destination host unknown		•
	8	source host isolated (obsolete)		•
	9	destination network adimistrativly prohibited		•
	10	destination host administratively prohibited		•
	11	network unreachable for TOS		•
	12	host unreachable for TOS		•
	13	communication administratively prohibited by filtering		•
	14	host precedence violation		•
	15	precedence cut-off in effect		•
4	0	source quench (elementary flow control)		•
5		redirect		•
	0	redirect for network		•
	1	redirect for host		•
	2	redirect for TOS and network		•
	3	redirect for TOS and host		•
8	0	echo request	•	
9	0	router advertisement	•	
10	0	router solicitation	•	
label 11		time exceeded		
	0	time-to-live equals 0 during transit		•
	1	time-to-live equals 0 during reassembly		•
12		parameter problem		
	0	IP header bad		•
	1	required option missing		•
13	0	timestamp request	•	
14	0	timestamp reply	•	
15	0	information request (obsolete)	•	
16	0	information reply (obsolete)	•	
17	0	address mask request	•	
18	0	address mask reply	•	

Table 1.3: ICMP messages

while the response messages are those labeled with reply.

1.8 Network basic commands in Linux

1.8.1 ifconfig

The command `ifconfig` (**i**nter**f**ace **config**uration) is used to configure network interfaces, for example:

- Check the IP address allocation from a machine interfaces
- Set the IP address and network mask of an interface
- Limit the value of the MTU of an interface
- Set the physical address of the interface if the interface supports this operation

The syntax for the command `ifconfig` varies depending on the operation to perform, but to assign IP addresses, the syntax is:

`ifconfig` *interface #IP* `netmask` *#mask* `broadcast` *IP_broadcast*

where interface is the interface name to be configured (eg eth1 or eth2), #IP is the IP address to be assigned to this interface (eg 192.168.1.1), `netmask` is part the syntax and indicates that the next argument is the network mask to be used (*#mask*), for example 255.255.255.0. Finally `broadcast` is a keyword that indicates that the next argument

is the IP broadcast address to be used in the interface (for example 192.168.1.255). As an example, the following command:

```
$ifconfig eth0 192.168.1.1 netmask 255.255.255.0 broadcast 192.168.1.255
```

Note. The symbol $ is the "prompt", it should not be typed on the console.

configures the eth0 interface with the IP address 192.168.1.1 with mask 255.255.255.0 and broadcast IP address 192.168.1.255.

The fact of assigning an IP address to an interface causes the operating system update its routing table, automatically adding the direct delivery route related to the assingning that has been carried out.

The command `ifconfig` can also disable an interface if it is not wanted to be used, for example:

```
$ifconfig eth0 down
```

In this case, the direct delivery route related to the configuration of eth0 is deleted from the routing table and so are all the indirect delivery routes that were using eth0 as output interface.

To set an interface up again with the previous configuration, the following command must be entered:

```
$ifconfig eth0 up
```

To see how the machine interfaces are configured machine, use the syntax:

`ifconfig [interface]`

where [interface] is optional. If any interface name is used, the command will return the configuration of all interfaces "up", whether or not they have an assigned IP. If we provide an interface name, the command will return the interface configuration. To check the configuration of all interfaces on a machine regardless of whether or not they are "up":

```
$ifconfig -a
```

A machine can reply in relation to the IP address of an interface, although it may be in down state. Sometimes this behavior causes problems in the laboratory, therefore, when a network interface is not needed, the most useful it is to remove the IP address assigned and then put the network interface down state. To remove the IP address to an interface what it is actually down is to assign the address 0.0.0.0. For example, to delete the IP address of the eth1 interface we would execute the following command:

```
$ifconfig eth1 0.0.0.0
```

Finally, using the ifconfig command we can also assign more than one IP address to an interface as follows:

```
$ ifconfig eth1 10.0.0.1/24
$ ifconfig eth1:0 10.0.0.2/24
$ ifconfig eth1:1 10.0.0.3/24
```

In the above example, the first command assigns the first direction (10.0.0.1) to the network card eth1. The second command assigns a second address (10.0.0.2), known as "0" alias to the network card eth1. Finally, the third command assigns a third direction (10.0.0.3), known as "1" alias to the network card eth1. In this way, IP addresses could be added successively to a network interface.

1.8.2 ARP

The command `arp` is used to query and manipulate the arp table on the computer. During the practical exercises we will use this command to know the contents of the arp table as follows:

```
$arp -an
```

If it is wanted to delete an entry from the arp table of our computer, we use the syntax:

```
$arp -d hostname
```

where hostname is the IP or the name (fqdn) of the computer we want to erase from our arp table.

1.8.3 route

El comando `route` sirve para consultar y manipular las tablas de rutas del sistema. Para consultar las tablas de rutas, el comando que se utiliza es: The command `route` is used to query and manipulate the routing tables of the system. To view the routing tables, the command used is:

```
$route -n
```

To explain the output of this command, we will revise an example (we only represent the fields that are useful during the practical exercises):

```
router1:~# route -n
Kernel IP routing table
Destination     Gateway         Genmask         Iface
10.0.0.0        0.0.0.0         255.255.255.0   eth1
192.168.1.0     0.0.0.0         255.255.255.0   eth2
0.0.0.0         192.168.1.2     0.0.0.0         eth2
```

The `Destination` field indicates the destination network address. The `gateway` field indicates through whom the delivery of the packet must be carried out. If the `gateway` fild value is 0.0.0.0, it indicates that delivery is direct, and if the `gateway` fild value is a host IP address of a host, it indicates that the package must be delivered to that host (indirect delivery) in order to reach the "`destination`" network. The `Genmask` field is the value of the mask to be used to calculate the IP addresses that belong to a certain network. For example, to the 10.0.0.0 network, it is used a Genmask of 255.255.255.0, which indicates that the range of IP addresses ranging from 10.0.0.0 to 10.0.0.255 are addresses that belong to the destination network. Finally the `Iface` field indicates through which interface the packet transmission will be performed.

The command `route` is used to add and delete routes, using the following syntax:

`route (add | del) -net #IP_net netmask #mask [gw #IP_router]`

For example, suppose we have the routing table before and we want to add an indirect delivery route to the network 192.168.100.0/24 through the router 10.0.0.5:

```
$route add -net 192.168.100.0 netmask 255.255.255.0 gw 10.0.0.5
```

If we later run the `route-n` command, we obtain:

```
router1:~# route -n
Kernel IP routing table
Destination     Gateway         Genmask         Iface
10.0.0.0        0.0.0.0         255.255.255.0   eth1
192.168.1.0     0.0.0.0         255.255.255.0   eth2
192.168.100.0   10.0.0.5        255.255.255.0   eth1
0.0.0.0         192.168.1.2     0.0.0.0         eth2
```

If we want to clear the path again, then we run the command:

```
$route del -net 192.168.100.0 netmask 255.255.255.0 gw 10.0.0.5
```

1.8.4 ping

Echo ICMP messages echo have been used traditionally to check the status of the connections between two network devices (hosts, routers, ...).

In the echo ICMP message, the type field has the values 0 or 8 (reply, request) and the field code has the value 0. The host sends an echo request ICMP message and it sets the identifier and sequence number fields to values that he chooses, and it also sets the content of the data. The host receiving the request responds with a reply message with the same values in the identifier and sequence number fields and with the same received content (that is, the host performs an "echo" of the received message).

The `ping` program is used to send an echo ICMP message request. The reply is handled directly in the kernel of the system that automatically sends the reply message. On Unix implementations, the identifier field is set to the PID value of the process that sends messages. This allows to `ping` to identify received replies in case there are more `ping` processes running.

The value of the sequence number field is initialized to 0 and it is incremented each new sent message, allowing to detect if a package is lost.

The `ping` program calculates the RTT (Round Trip Time)), the time between when an echo request is sent and received the reply.